

VaxPlan

End-User Guide

A role-by-role manual for the VaxPlan GIS microplanning platform.

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A practical, role-by-role manual for the VaxPlan GIS microplanning platform. Use the table of contents to jump to your role.

environment, your tenant administrator may have customised the labels (for example "Province" ! "Region") — the workflows below are unchanged.

- 10. What VaxPlan can do
 - 10.1.1. Plan a dance
 - 10.1.2. Turn the screen and switching country
 - 10.1.3. Turn off a village microplan
 - 10.1.4. Add a session in the field
 - 10.1.5. Add a session in the field
 - 10.1.6. Add a session in the field
 - 10.1.7. Online mode and sync
 - 10.1.8. Supply
- 10. District managers for review and oversight
- 10.1. District coordinators – approvals and visibility
- 10.2. Tenant onboarding (New Ministry of Health)
- 11. Map and boundary management
- 12. Settlement intelligence and zero-dose targeting
- 13. Settings, customisation, and labels
- 14. Supervision visits
- 15. Reports and exports
- 16. Troubleshooting
- Glossary

VaxPlan is a multi-country microplanning system used by Ministries of Health to plan, run, and track routine immunisation and supplementary immunisation activities (SIAs).

- Microplanning:** facility-level quarterly plans that combine catchment population, vaccine schedules, and outreach intent into a list of executable sessions.
- Coverage analytics:** by antigen, by dose, by month, by location, with under-immunised and zero-dose surfacing.
- Stock and supply:** vaccine requirements, wastage thresholds, cold-chain stock balances, and reconciliation.
- Mapping:** every facility, village, settlement, and session plotted on Leaflet maps with admin boundary overlays (GeoBoundaries + custom GeoJSON uploads).
- Multitenant SaaS:** each Ministry of Health is a separate tenant with its own data, users, and SSO. Users with cross-tenant access can switch between countries from the header.
- Where the data lives.** Everything is stored in PostgreSQL on a tenant-isolated schema. Facility and village reference data is loaded once during onboarding and then maintained by national administrators. Day-to-day operational data (sessions, coverage, stock) is written by facility staff.

Role	What they can do	Where they work
Facility clerk	Authors microplans and sessions, captures session results, manages stock balances.	Their own facility only.
Facility in-charge	Same as clerk, plus signs off (submits) the microplan and session results.	Their own facility only.
District manager	Reviews and approves microplans from facilities in the district, runs supervision visits, reads coverage.	Their district.
Provincial coordinator	Approves district-level plans, sees rolled-up coverage, escalates issues.	Their province.
National admin	Manages users, facilities, vaccine schedule, labels, boundaries, and the country dashboard.	Whole country (tenant).
Tenant superadmin	Onboards new tenants, configures SSO, and provisions the first national	

admin.

Microplan authoring (creating new microplans and session plans) is **restricted to facility staff** (clerk and in-charge). Higher roles are reviewers and approvers; they cannot author plans on behalf of a facility. This keeps accountability with the people who actually run the sessions.

3. Signing in

VaxPlan supports two sign-in modes:

Replit sign-in (used in the demo environment and during onboarding). **Tenant SSO** — once your Ministry of Health is fully onboarded, you sign in with your own organisational identity (OIDC or SAML — for example Microsoft Entra, Google Workspace, Okta). **To sign in** — Open the VaxPlan URL provided by your administrator (each tenant has either a subdomain or a path-based URL). **Work Sign in** — you are redirected to your identity provider; complete the login there. On first login, your home tenant is set automatically from your email domain (if your administrator has configured a domain mapping) or from the signup invite you accepted. You'll land on your home tenant's dashboard. If you signed up but no role has been granted yet, you'll see a "pending approval" message. Your district or national administrator needs to confirm your role before you can use the system. They will receive an inbox notification automatically.

4. The home screen and switching country

The header has three constant elements:

Tenant switcher (top-left) — shows your current country. If you have access to more than one tenant (rare, mostly used by cross-MoH regional coordinators), you'll see a dropdown to switch. Switching does not move your home — it only changes what data you're viewing.

Navigation sidebar (collapsible) — your modules. Items that your role can't access are hidden, so the menu adapts to who you are.

Profile menu (top-right) — language, theme (light/dark), and sign-out.

Cross-tenant rule: when you switch to another country, you become a read-only viewer of that tenant's data. The system rejects any write you try to perform outside your home tenant with a clear error message. This is intentional — it lets a regional advisor see neighbouring countries' plans without risking accidental edits.

5. Facility staff — your daily workflow

This is the most important section for clerks and in-charges. Read it end-to-end the first time, then return to specific subsections as needed. There is also a separate one-page card, [QUICKSTART_FACILITY.md](#), you can print and pin next to your workstation.

5.1 Build a routine microplan

A **microplan** is your facility's quarterly plan. It declares:

the target population (children 1-5 and pregnant women by default — configurable per tenant), the outreach sessions you intend to run (fixed-site sessions are automatic).

Click the sidebar, open **Microplans > Routine**.

Step 1 — Scope. Pick the quarter and year. Your facility is pre-filled from your profile and cannot be changed.

Step 2 — Catchment. Tick the villages the facility will serve this quarter. The list is your facility's assigned villages from the registry; if a village is missing, ask your national admin to add it before continuing.

Step 3 — Register target population. On the left, the wizard adds default targets. Three data sources feed this:

Manual override (with a justification note). Pick the source per village; the wizard sums everything and shows you the totals.

Step 4 — Vaccine schedule. The default schedule is your tenant's. Untick antigens that don't apply (for example, if your facility doesn't carry HPV).

Step 5 — Outreach intent. Declare how many outreach sessions per village you expect to run. The system creates one session plan per (village x month x declared count). You can edit individual sessions later.

Step 6 — Review and submit. Check the totals, then **Save as draft** (you can keep editing) or **Submit for approval** (your district manager sees it in their queue).

Tip. You can save a draft at any step. Drafts are private to you until you submit.

5.2 Plan a session

Once a microplan is approved, its sessions appear on the **Sessions** page. Each session is created automatically from the microplan's outreach intent. You can also add ad-hoc sessions for defaulter follow-up.

Open **Sessions** from the sidebar.

Use the **Province > District > Facility** cascade filter at the top to find your sessions. The row count below changes as you filter.

Click a session name. The edit dialog opens.

Scheduled date — the day you'll run it.

Cold chain plan — vaccine carrier, ice packs, expected vaccines.

Click **Save**.

If GPS coordinates are missing for the village, the system warns you when you save and offers a "Capture GPS now" link to record them from your phone in the field.

5.3 Run a session in the field

The session execution screen is designed for use **offline**, on a phone or tablet, while you're at the village.

For each session, the screen prompts for capture mode.

Confirm antigens administered: The system auto-picks the next due dose based on the schedule.

The session totals update live. Stock balances on the device decrement automatically.

You can capture an entire day's session with no connectivity. The device queues every entry into an **offline outbox** (see 5.7).

5.4 Mark a session done

After you've finished vaccinating:

Confirm the per-antigen counts: The system pre-fills these from your capture; you can adjust if your physical tally differs.

Add session notes (issues, no-shows, supply problems).

Click **Submit**.

What happens behind the scenes:

Per-antigen counts are validated against your tenant's vaccine schedule. Known codes are stored under their canonical name (so **OPV-1** and **OPV-1** are treated the same).

Unknown codes (usually from older offline entries) are stored in a separate bucket so they still count toward totals but don't pollute per-antigen rollups. You'll see a warning if any were found, and your national

admin can review them in the audit log.

The session is locked. Reopening requires district approval.

5.5 Coverage and the under-immunised list

Coverage is shown on the **Coverage** page. You'll see:

A breakdown by village, by coverage date to date.

An **under-immunised list** of children who have started but not completed a vaccine series (for example, OPV-1 done but OPV-2 missing past the due date).

Acting on the under-immunised list:

Click **Create defaulter follow-up session** to spin up a new outreach session targeting that child's village.

The session is tagged so it shows up under the **Defaulter follow-up only** filter on the Sessions page.

5.6 Stock, wastage, and supply

The **Stock** page tracks vaccine balances at your facility:

Reorders by auto-debit and per-vial from a district order; the delivery note.

Wastage — auto-computed from session counts vs. opened vials, with a per-antigen wastage threshold.

Vials wasted above threshold trigger an alert visible to your in-charge and district manager.

The **monthly stock summary** is your end-of-month return: review, adjust if you find a discrepancy on physical count, and submit.

5.7 Offline mode and sync

VaxPlan works without an internet connection. Here's what you need to know:

The first time you sign in, the app **caches your reference data** (facilities,

villages, vaccine schedule, microplans) into an on-device IndexedDB.

When you create or update something offline (a session result, a stock movement, a new defaulter session), it goes into an **outbox**. The header shows a small cloud icon with the number of pending items.

If a sync entry is rejected (for example, a session was already closed on the server), the system shows the

rejection inline and asks you to resolve it.

Best practice. Sync at the end of each session day, when you're back in cellular range. Don't let the outbox grow longer than a week's worth of entries.

6. District managers — review and oversight

You sit between facilities and the province. Your day-to-day:

Approval queue. Open **Approvals**. You'll see microplans submitted by facilities in your district. For each,

you can:

Approve — the plan locks and its sessions go live.

Reject — for plans that should be rebuilt from scratch.

Coverage rollup. The **Coverage** page shows you the whole district at a glance. Drill down by facility or

village.

Stock alerts. You'll receive a weekly digest of facilities with stockouts, wastage above threshold, or

upcoming expiries.

Cross-facility intelligence. The **Map** view shows every session in your district pinned by status (planned,

conducted, overdue, cancelled). Use it to spot uneven coverage by location.

You **cannot** author microplans for a facility — that responsibility stays with facility staff. You can, however,

edit catchment assignments (which villages belong to which facility) if you spot a boundary issue.

7. Provincial coordinators — approvals and visibility

Your role mirrors the district manager's, scoped to the province:

District-level plan approvals. When a district manager signs off on aggregated district-level outreach plans

(for SIAs, mostly), they come to you next.

Resource allocation. See which districts are on track and which are slipping.

Supply request workflow. You also have access to the **National admin** read-only views (you cannot edit users or facilities, but you can

see them).

National admins are the power users for your country. Your modules:

- Users.** Invite users, assign roles, suspend or reactivate accounts. You can also bulk-import users from a CSV.
- Facilities.** The registry of all facilities. Import from CSV (a template is downloadable), edit GPS coordinates, merge duplicates or retire facilities.
- Vaccine schedule.** Your tenant's authoritative schedule. Adding an antigen here makes it available in microplans nationwide.
- Country dashboard.** Top-line KPIs for the country, including coverage by antigen, dropout rates, stock health, and supervision compliance.
- Additional escalation rules.** A district or province rejected escalates to you.

National admins can also configure **scheduled jobs** — population refresh from WorldPop, stock-alert digests, and supervision digests all run on schedules you can tune in **Settings ! Schedules**

This section is for the VaxPlan superadmin onboarding a new country.

- Create the tenants and locations.** Tenants! New and pick:
 - Default public agency labels (e.g. Province/District/Facility or Region/State/Health Area).
 - Default vaccine schedule (clone from a sibling country or you have one, then edit).
- Configure SSO.** Add the OIDC or SAML configuration for the ministry's identity provider. Test the connection before going live.
- Map email domains.** Adding `@health.gov.xx` makes anyone who signs in from that domain land on this tenant by default.
- Onboard the first national admin.** They will receive an invite email and be able to onboard everyone else.
 - Local admin onboarding:** use the **Boundary Manager** (\$10).
 - Facilities onboarding via CSV:** Import via CSV.
 - Population** — either ingest a WorldPop raster (national admin can do this on demand) or rely on registered population.
- Set the approval workflow.** Decide whether plans need 1, 2, or 3 levels of approval (facility! ' district! ' province! national).
- Go live.** The National admin sends out user invites and training links.

Every map in VaxPlan (Sessions, Coverage, Settlement intelligence, Microplans) draws boundaries on top of OpenStreetMap tiles. Boundaries come from two sources:

- GeoBoundaries API** — public, covers 200+ countries, available for admin levels 0 to 2 or 3 depending on the country.
- Custom GeoJSON upload** — your own files, for levels GeoBoundaries doesn't cover (e.g. South Sudan Payam) or for your authoritative national geometry.

To fetch from GeoBoundaries:

- Click **Set Admin Boundary Manager**.
- Click **Fetch Boundaries**. Large countries (Nigeria, DRC, Ethiopia) take 30 to 60 seconds.

To upload custom GeoJSON load Custom GeoJSON.

Use a bounding box code (see below). Files up to 50 MB are accepted.

Click **Upload & Store**.

GADM users: GADM ships shapefiles, not GeoJSON. Convert with the free mapshaper.org website (drag in the .shp, .shx, .dbf files, export as GeoJSON).

For countries where village-level registration is patchy (parts of South Sudan, PNG highlands, Sahel), VaxPlan offers a **settlement intelligence** layer. It overlays:
 Travel-time isochrones from each health facility.
 The **zero-dose map** uses this data to highlight settlements with no recorded vaccinations. Click a hotspot to:
 See the address, search in an atlas, or view the map.
 Create an outreach session targeting the hotspot.
 This module is most useful for the **district manager** and **provincial coordinator** roles when planning quarterly microplanning calendars.

Admin level labels: Each tenant can rename the four hierarchy levels. The default is Country / Province / District / Facility. Common alternatives:
 Zambia: Country / Province / District / Constituency
 Portugal: Municipality / District / Parish / Freguesia
 Brazil: Country / State / Municipality / City / Neighborhood
 You can also use the default language labels on the header and on all PDF exports

Language: Choose one default language for all users in your tenant. English, French, and Portuguese are bundled; ask the VayPlan team for additions; mark which are routine vs. campaign

Wastage thresholds: Per antigen: What percentage of doses was wasted in doses at alert.

Supervision is a first-class workflow:

Schedule visits. District managers open **Supervision I: New visit**, pick a facility, a date, and a supervisor. **Visit checklist.** When the supervisor arrives, they open the visit on their phone and tick the checklist (cold chain, records, stock, session register, etc.). **Supervision I: Generate** item becomes an action item with an owner and a due date. **Submit.** The facility in charge to signs the report, both get a copy. The **Supervision digest** (a weekly summary) rolls up overdue actions to the district and provincial dashboards.

14. Reports and exports

Most tables in VaxPlan have an **Export** button that produces an Excel workbook with the currently filtered rows.

For more formal outputs, use **Reports I: Generate** (ask a national admin) or your browser blocked location/tile fetches. Try a different browser or hard-refresh. **Supervision report (PDF, per visit)**. **Stock and wastage report (Excel, per facility, per month)**. All reports honour the geo filters you've selected on the page.

15. Troubleshooting

I can't see my facility's sessions on the Sessions page.

Check the Province / District / Facility filter at the top of the page — if any are set, only matching sessions are shown. Clear them to see everything you're allowed to see.

The map is blank. Either you don't have boundaries loaded for the level you're viewing (ask a national admin) or your browser blocked location/tile fetches. Try a different browser or hard-refresh.

413 Request Entity Too Large When uploading a boundary.

That used to happen for files over 100 KB. Files up to 50 MB are now accepted. If you see this on a smaller file, the file may not be valid GeoJSON; try opening it in geojson.io to validate.

GeoBoundaries has no ADM3 boundary error.

GeoBoundaries doesn't publish every admin level for every country. For South Sudan, only ADM0-ADM2 are upstream. For Pavam, you need to upload a custom GeoJSON (OCHA HDX is a good source).

My country code is rejected with "must contain exactly 3 characters". Use the ISO 3166-1 alpha-3 code (e.g. **SSD** for South Sudan, **ZMB** for Zambia, **PNG** for Papua New Guinea, **KEN** for Kenya). The 2-letter alpha-2 codes (**SS**, **ZM**) are not accepted.

I marked a session done but it shows zero coverage.

The per-antigen counts may use unknown codes (older offline outbox entries). Open the session, check the "unmapped antigens" warning, and ask your national admin to standardise the codes via the audit log workflow.

Sync failed for some outbox entries.

Tap the cloud icon to see

which ones. Most failures are because the underlying session was closed or deleted on the server. Reopen the entry, resolve the conflict, and retry.

I'm logged in but I see "pending approval".

A national or

district admin needs to confirm your role. Contact your administrator; they will see the request in their inbox.

16. Glossary

Antigen — A vaccine type (BCG, OPV, Penta, MCV1, etc.).

Coverage — The set of villages and facilities that received a given dose, in a given period.

Denominator — A child was status (either coverage or intended outreach).

Dropout — The percentage of children who received an earlier dose but did not receive a later one (e.g. Penta1, Penta3).

Outreach session — A vaccination session held at the facility.

Outreach session — A session held away from the facility, usually in a home.

Supervision I — A weekly report (country) on the VaxPlan dashboard. Each report has isolated data.

Zero-dose child — A child of vaccination age who has received no doses of any vaccine.

If you spot an error in this guide or want a topic added, ask your national admin to file an issue with the VaxPlan team. The guide is versioned alongside the application code.

